

Renal Reference — AKI Diagnostic Steps

one clean AKI diagnostic steps table

Step	Main question	What to check	What it tells you
1 Confirm AKI	Is this truly AKI?	↑ SCr ≥0.3 mg/dL in 48h, or ≥1.5× baseline in 7 days, or UO <0.5 mL/kg/h for >6h	Confirms AKI by KDIGO criteria
2 History + physical	Is there an obvious cause?	Vomiting, diarrhea, bleeding, HF, cirrhosis, NSAIDs, ACE-I/ARB, contrast, sepsis, BPH symptoms	Points toward prerenal, intrinsic, or postrenal
3 Urine output pattern	Is the patient oliguric, anuric, or polyuric?	Oliguria <400 mL/day, anuria, or normal/high urine output	Oliguria = severe AKI; anuria suggests obstruction/vascular catastrophe; polyuria can occur in partial obstruction or tubular injury
4 Rule out obstruction	Is urine flow blocked?	Bladder scan, renal ultrasound, Foley obstruction, hydronephrosis	If obstruction → postrenal AKI
5 Urine sediment	Where is the injury?	Hyaline casts, muddy brown casts, RBC casts, WBC casts, crystals	Best clue for type of AKI
6 Renal failure indices	Do labs support the pattern?	BUN/Cr, FeNa, FeUrea, urine Na, urine osmolality	Helps separate prerenal vs ATN, but has exceptions
7 Interpret BUN/Cr	Is urea rising more than creatinine?	BUN/Cr >20:1	Suggests prerenal AKI because kidney reabsorbs urea with water
8 Interpret FeNa / FeUrea	Is kidney conserving sodium/urea?	FeNa <1%, urine Na <20, FeUrea <35%	Supports prerenal AKI • is there FeNa exception cases - that are actually intrensic (sepsis/contrst/rhabdomylsis/early GN)
9 Look for ATN pattern	Are tubules damaged?	FeNa >1%, muddy brown casts, urine osm <350, BUN/Cr ~10–15	Suggests intrinsic AKI / ATN
10 Check exceptions	Could indices mislead you?	Diuretics, CKD, contrast, rhabdo, early sepsis, GN, GI bleed, steroids	Prevents board traps

1. Confirm AKI

WHAT YOU SEE

Step 1 row: 'Is this truly AKI?'

WHAT IT ENCODES

Apply KDIGO criteria: $\uparrow$ SCr ≥ 0.3 mg/dL within 48h, OR SCr $\geq 1.5\times$ baseline within the prior 7 days, OR urine output <0.5 mL/kg/h for >6 h. Stage by severity — Stage 1: Cr 1.5 - $1.9\times$ baseline or ≥ 0.3 rise; Stage 2: 2 - $2.9\times$; Stage 3: $\geq 3\times$ baseline, or Cr ≥ 4.0 , or RRT initiation, or UO <0.3 mL/kg/h for ≥ 24 h / anuria ≥ 12 h.

BOARD TRAP

WRONG answer is diagnosing AKI from one elevated creatinine with no baseline and no time course. A chronically elevated, stable Cr is CKD. Discriminator: AKI demands an acute CHANGE (≥ 0.3 in 48h or $\geq 1.5\times$ in 7 days); small echogenic kidneys, anemia, and renal osteodystrophy on labs point toward CKD instead.

2. History + physical

WHAT YOU SEE

Step 2: obvious cause?

WHAT IT ENCODES

Target the history to the three categories: vomiting/diarrhea/bleeding/poor intake/diuretics $\rightarrow$ prerenal volume loss; HF/cirrhosis/sepsis $\rightarrow$ reduced effective volume; new NSAIDs/ACE-I-ARB/contrast/aminoglycosides $\rightarrow$ drug-induced; rash/arthralgia/hemoptysis $\rightarrow$ GN/vasculitis; BPH symptoms/anuria/distended bladder $\rightarrow$ postrenal. Exam: volume status (JVP, mucous membranes, edema), bladder distension, livedo/blue toes, rash.

BOARD TRAP

WRONG answer is anchoring on the most recent admission diagnosis and skipping the medication reconciliation. Discriminator: the 'triple whammy' (NSAID + ACE/ARB + diuretic) and recently started PPIs/antibiotics are common, reversible precipitants that only a careful med history reveals — review the full drug list and timing relative to the Cr rise.

3. Urine output pattern

WHAT YOU SEE

Step 3: oliguric, anuric, or polyuric?

WHAT IT ENCODES

Oliguria = UO <400 - 500 mL/day (or <0.5 mL/kg/h); anuria = <100 mL/day. ANURIA is severe and narrows the differential to complete obstruction, a vascular catastrophe (bilateral renal artery/vein occlusion), severe cortical necrosis, or RPGN. Nonoliguric AKI (common in aminoglycoside/contrast ATN) carries a better prognosis. Polyuria can reflect partial obstruction, the recovery phase of ATN, or post-obstructive diuresis.

BOARD TRAP

Polyuria rules out AKI — this is the WRONG answer. High urine volume occurs in the RECOVERY phase of ATN, in partial obstruction, and as post-obstructive diuresis, all of which are AKI states. Discriminator: judge AKI by creatinine/UOP per KDIGO, not by urine volume alone — a polyuric patient can still have a rising or non-clearing Cr.

4. Rule out obstruction

WHAT YOU SEE

Step 4: is urine flow blocked?

WHAT IT ENCODES

Exclude postrenal early because it is rapidly reversible. Steps: post-void bladder scan (or place a Foley to relieve outlet obstruction and measure residual), then renal ULTRASOUND for hydronephrosis. Relief is often curative; anticipate post-obstructive diuresis (replace volume and monitor K, Na, Mg, phosphate).

BOARD TRAP

WRONG answer is excluding obstruction solely on a renal ultrasound with no hydronephrosis. Early, dehydrated, or encased obstruction (retroperitoneal fibrosis, pelvic tumor encasing ureters) may show no hydronephrosis. Discriminator: if suspicion is high (anuria, known pelvic malignancy, distended bladder), proceed to CT or retrograde pyelography despite a 'normal' US.

5. Urine sediment

WHAT YOU SEE

Step 5: where is the injury?

WHAT IT ENCODES

Microscopy is the single best bedside clue to AKI TYPE: bland/hyaline casts = prerenal; muddy-brown granular + renal tubular epithelial cell casts = ATN; RBC casts/dysmorphic RBCs = glomerulonephritis; WBC casts + eosinophiluria = AIN; crystals = ethylene glycol (oxalate), uric acid (tumor lysis), or drug crystals (acyclovir, sulfa, indinavir).

BOARD TRAP

WRONG answer is treating isolated hematuria as automatically glomerular. Isomorphic RBCs with clots and dysuria point to a urologic/lower-tract source (stone, tumor, infection). Discriminator: RBC CASTS and DYSMORPHIC RBCs localize the bleeding to the glomerulus — without casts, pursue a urologic workup (cystoscopy/CT urogram) instead of a renal biopsy.

6. Renal failure indices

WHAT YOU SEE

Step 6: do labs support the pattern?

WHAT IT ENCODES

Quantitative indices to separate prerenal from ATN: BUN:Cr, FENa = $(\text{UNa} \times \text{PCr}) / (\text{PNa} \times \text{UCr}) \times 100$, FeUrea, urine Na, and urine osmolality. Prerenal: FENa $<1\%$, urine Na <20 , urine osm >500 , BUN:Cr >20 . ATN: FENa $>2\%$, urine Na >40 , urine osm <350 , BUN:Cr 10-15.

BOARD TRAP

Indices are definitive — this is the WRONG answer. FENa fails in many real scenarios: it is LOW in early sepsis, contrast nephropathy, pigment nephropathy (rhabdo/hemolysis), early GN, and HRS; and it is FALSELY HIGH in patients on diuretics or with CKD. Discriminator: never diagnose off FENa alone — combine it with the urine sediment and clinical context, and use FeUrea ($<35\%$ prerenal) when the patient is on diuretics.

7. Interpret BUN/Cr

WHAT YOU SEE

Step 7: BUN/Cr >20:1

WHAT IT ENCODES

A BUN:Cr ratio >20:1 suggests prerenal AKI: low tubular flow lets the nephron reabsorb urea (along with avidly reabsorbed Na and water) while creatinine is not reabsorbed, so BUN rises out of proportion. In ATN the ratio falls toward 10-15:1 because tubular reabsorption is impaired.

BOARD TRAP

WRONG answer is reading every BUN:Cr >20 as prerenal. GI BLEEDING (urea load from digested blood), high-protein/TPN feeding, corticosteroids, and tetracyclines all raise BUN independent of perfusion. Discriminator: cross-check with FENa <1%, bland sediment, and volume status — and look for melena/hematemesis explaining an extrarenal BUN bump.

8. Interpret FENa

WHAT YOU SEE

Step 8: FENa <1%, urine Na <20, FeUrea <35%

WHAT IT ENCODES

FENa <1% (with urine Na <20 and FeUrea <35%) supports prerenal AKI — the RAAS/ADH-activated nephron conserves sodium. FeUrea is preferred when the patient is on loop/thiazide diuretics, which artificially raise urinary Na and FENa (FeUrea <35% still indicates prerenal under diuretics).

BOARD TRAP

Low FENa = always prerenal — this is the WRONG answer. The classic low-FENa EXCEPTIONS are intrinsic AKI: early sepsis, contrast nephropathy, pigment nephropathy (rhabdomyolysis/hemolysis), early/acute GN, and hepatorenal syndrome. Discriminator: when a low FENa accompanies muddy-brown casts, CK elevation, recent contrast, or RBC casts, the lesion is intrinsic despite the 'prerenal' index.

9. ATN pattern

WHAT YOU SEE

Step 9: tubules damaged?

WHAT IT ENCODES

Intrinsic ATN profile: FENa >2%, urine Na >40, urine osm <350 (isosthenuric — lost concentrating ability), BUN:Cr ~10-15, and MUDDY-BROWN granular / renal tubular epithelial cell casts on microscopy. Clinical course runs through initiation, maintenance (oliguric, where complications peak), and recovery (polyuric) phases.

BOARD TRAP

WRONG answer is expecting ATN to bounce back quickly with a fluid bolus like prerenal AKI does. Established ATN recovers over days to weeks and does not respond promptly to fluids; overfilling causes pulmonary edema. Discriminator: muddy-brown casts + FENa >2% + isosthenuria = ATN (supportive care, avoid nephrotoxins, dose-adjust drugs); bland urine + FENa <1% = the volume-responsive prerenal state.

10. Check exceptions

WHAT YOU SEE

Step 10: could indices mislead?

WHAT IT ENCODES

Before committing, ask what could distort the indices: DIURETICS and CKD falsely RAISE FENa (use FeUrea); contrast, rhabdo/pigment, early sepsis, early GN, and HRS falsely LOWER FENa; GI bleed/steroids/high protein raise BUN:Cr without prerenal physiology. Reconciling the index with the sediment and history is what prevents the board trap.

BOARD TRAP

WRONG answer is reporting FENa without noting the patient just received furosemide. Diuretics force natriuresis and push FENa >1% even in a volume-depleted, prerenal patient. Discriminator: in any diuretic-treated patient substitute FeUrea (<35% = prerenal); a FENa >1% under active diuresis does NOT establish ATN.

11. What to check column

WHAT YOU SEE

Middle column header 'What to check'

WHAT IT ENCODES

This column pairs each diagnostic question with the specific test or finding to obtain — e.g. confirm AKI → trend creatinine and urine output against baseline; localize injury → urine microscopy; separate prerenal vs ATN → FENa/FeUrea, urine Na, urine osm; exclude postrenal → bladder scan + renal ultrasound.

BOARD TRAP

WRONG answer is ordering a renal biopsy early for undifferentiated AKI. Biopsy is reserved for cases where noninvasive workup is inconclusive and intrinsic disease (GN/vasculitis, AIN not improving, unexplained AKI) is suspected. Discriminator: most AKI is diagnosed by history, urinalysis/sediment, indices, and ultrasound — biopsy comes only after these point to a treatable parenchymal lesion.

12. What it tells you

WHAT YOU SEE

Right column 'What it tells you'

WHAT IT ENCODES

This column maps each finding to its diagnostic meaning: KDIGO change confirms AKI and stages it; sediment localizes the compartment (glomerular/tubular/interstitial); indices and BUN:Cr separate prerenal from ATN (with exceptions); ultrasound/bladder scan confirm or exclude obstruction. The synthesis directs therapy — fluids, drug withdrawal, immunosuppression, or relief of obstruction.

BOARD TRAP

WRONG answer is acting on a single finding in isolation (e.g., treating low FENa as prerenal and bolusing fluids) without integrating the column. Discriminator: AKI diagnosis is a CONVERGENCE of history + sediment + indices + imaging — when one clue conflicts with the others (e.g., low FENa but muddy-brown casts), trust the more specific sediment and the clinical picture.